

COMP 699 – Course Project Proposal

Project Type: Software Development Project

Name of Project: AI PDF Assistant

Why an organization would find this system useful:

The AI PDF Assistant will help researchers and students quickly find and understand information in academic research papers. Users can upload academic PDF documents and ask questions in natural language instead of manually searching through lengthy papers.

Description of organizational requirements of the system:

The system will use React and Tailwind CSS for the frontend and Python with FastAPI for the backend. It will use PDF text extraction, embeddings, a vector database, LangChain, Retrieval-Augmented Generation (RAG), and a Large Language Model (LLM) to retrieve relevant information and generate answers.

Expected Advantages:

The system can reduce the time needed to search within research papers, make information easier to access, and provide relevant answers based on the uploaded documents. The initial implementation will focus on academic research papers, making the project realistic and manageable to complete within 14 weeks.

Special Requirements/Resources:

The project will require a computer with Python, React, FastAPI, LangChain, an embeddings model, a vector database, and an LLM. A local LLM such as Ollama may be used during development.